



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

ME2003-1 – MANUFACTURING TECHNOLOGY
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The point at which the cutting tool reaches a stage, beyond which it will not function satisfactorily until it is reground, is called as

A) Tool wear	B) Tool failure
C) Tool diffusion	D) Tool hardening
2. Which one of the following sets of forces are encountered by a lathe parting tool while grove cutting.

A) Tangential, radial; and axial	B) Tangential and radial
C) Tangential and axial	D) Radial and axial
3. The surface of the single point cutting tool on which the chips formed in cutting operation slide is called as

A) Flank	B) Heel
C) Rake Face	D) Shank
4. 9 Speed all geared head stock is used in lathe

A) To increase time	B) To decrease time
C) To change the direction of speed	D) To vary speed
5. In lathe machine the workpiece mainly

A) Reciprocates	B) Rotates
C) Oscillates	D) Remains stationery
6. The example for production lathe is

A) Engine Lathe	B) Tool room lathe
C) Turret Lathe	D) Speed lathe
7. The following is not a taper turning method in lathe

A) Swiveling the compound rest	B) Indexing the turret
C) Offsetting the tool post	D) Using a taper turning attachment
8. Lathe bed is made of

A) High speed steel	B) High carbon steel
C) Mild steel	D) Cast iron
9. Which of the following process is carried out to provide seating for head of screw?

A) Counterboring	B) Countersinking
C) Tapping	D) Reaming
10. In a Shaper machine the ram holding the tool gets _____ movement.

A) Rotary	B) Reciprocating
C) Both rotary and reciprocating	D) Oscillating
11. To reduce the total cutting time, the standard shaper is designed to complete return stroke _____

A) Slowly	B) Faster
C) At moderate speed	D) Intermittent
12. Which of the following operation is carried out for cutting internal threads?

A) Drilling	B) Tapping
C) Boring	D) Reaming

13. In centerless grinding, workpiece is supported by
 A) Centers B) Chuck
 C) Work rest D) Surface table
14. Following is not a type of centerless grinding
 A) Through feed grinding B) In feed grinding
 C) Internal centerless grinding D) Surface grinding
15. Which of the following is a surface finishing operation?
 A) Drilling B) Honing
 C) Milling D) Turning
16. Which of the following process have the lowest metal removal rate?
 A) Drilling B) Reaming
 C) Milling D) Honing
17. In Ultrasonic machining, the function of transducer is to
 A) Convert mechanical energy into heat B) Convert electrical energy into heat
 C) Convert electrical energy into mechanical vibrations D) Convert mechanical energy into electrical energy
18. In Electrical Discharge Machining (EDM) process the metal removal is carried out by
 A) Electrolysis B) Melting and vaporization
 C) Fracture of work material due to impact of grains D) Liquefying
19. In Abrasive jet machining, work piece material is removed by which of the following means?
 A) Vaporization B) Electro plating
 C) Mechanical abrasion D) Corrosion
20. In which of the following methods, an electrolyte is used?
 A) Ultrasonic Machining B) Electrochemical Machining
 C) Abrasive Jet Machining D) Laser Beam Machining

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Briefly explain the various cutting tool materials used in machining.	8	1	1	1
b)	List the differences between the Capstan and Turret lathe.	8	1	2	1
2. a)	Derive a relationship for shear angle in terms of rake angle and chip thickness ratio.	8	1	1	1
b)	Explain with figure the Bar Feeding mechanism used in Lathe machine.	8	1	2	1
3. a)	During orthogonal cutting using a single point cutting tool the following data was obtained. Cutting force $F_H = 1250\text{N}$, Feed force $F_V = 400\text{N}$, Rake angle $\alpha = 10^\circ$, Undeformed chip thickness = 0.2mm, width of cut = 2.5mm, chip thickness = 0.25mm, cutting speed = 150m/min. Determine (i) Chip thickness ratio, (ii) Shear angle (iii) Shear stress.	8	3	1	1
b)	Explain with figure the 9 speed All Geared Headstock Drive used in Lathe machine. Also list all the speed ratios obtained in terms of number of gear teeth.	8	1	2	1
Unit – II					
4. a)	Mention how the milling machines are classified. Explain with neat sketch the construction and working of Column and knee type Horizontal Milling machine.	8	1	3	1
b)	Explain the basic principle of Grinding operation. Also list the various factors affecting the selection of grinding wheels.	8	1	4	1
5. a)	Explain with diagrams the following Drilling machine operations. (i) Reaming (ii) Counter Boring (iii) Counter sinking (iv) Tapping.	8	1	3	1
b)	List the different types of Grinding operations. Explain the working of Cylindrical Centerless Grinding operation with a neat sketch.	8	1	4	1

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| 6. a) | Draw the sketch of a Twist Drill showing all the terminologies involved. Also draw the sketch of a Plain Milling cutter showing all its elements. | 8 | 1 | 3 | 1 |
| b) | Write short notes on: | | | | |
| | (i) Honing operation | | | | |
| | (ii) Lapping operation. | 8 | 1 | 4 | 1 |

Unit – III

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|-------|---|---|---|---|---|
| 7. a) | Describe the process of ECM with a sketch. Also list its advantages, disadvantages, and applications. | 8 | 1 | 5 | 1 |
| b) | Elaborate the process of AJM with a neat sketch and mention its merits, demerits and uses. | 8 | 1 | 5 | 1 |
| 8. a) | Explain EDM process with necessary figures and list its process parameters, Applications and Limitations. | 8 | 1 | 5 | 1 |
| b) | With a neat sketch describe the process of LBM mentioning its advantages, disadvantages and uses. | 8 | 1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
